

Railway Booking System

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Agenda

1. Project overview and current status
2. High-level system architecture and data flow
3. Main user workflows:
 - Account
 - Search
 - booking
 - manage bookings
4. Admin dashboard features planned and mockups

Flow diagram



Project Overview



Purpose

Provide an intuitive web interface for finding trains, selecting seats and booking tickets.



Core goals

Speed, clarity, data integrity and user session management.



Implemented features

User registration/login, searching trains, booking flow, booking history



Tems pending

Admin APIs, email confirmations, real payment gateway integration.

High-level Architecture



Frontend

static HTML/JS files (index.html, search.html, booking.html, register.html) served as UI



Backend

Spring Boot REST APIs handling authentication, trains, coaches, seats and bookings



Database

Stores JPA entities for User, Train, Station, Coach, Seat, Schedule and Booking



Data flow

User action → frontend request → backend processing → database persistence → frontend confirmation.

Browser



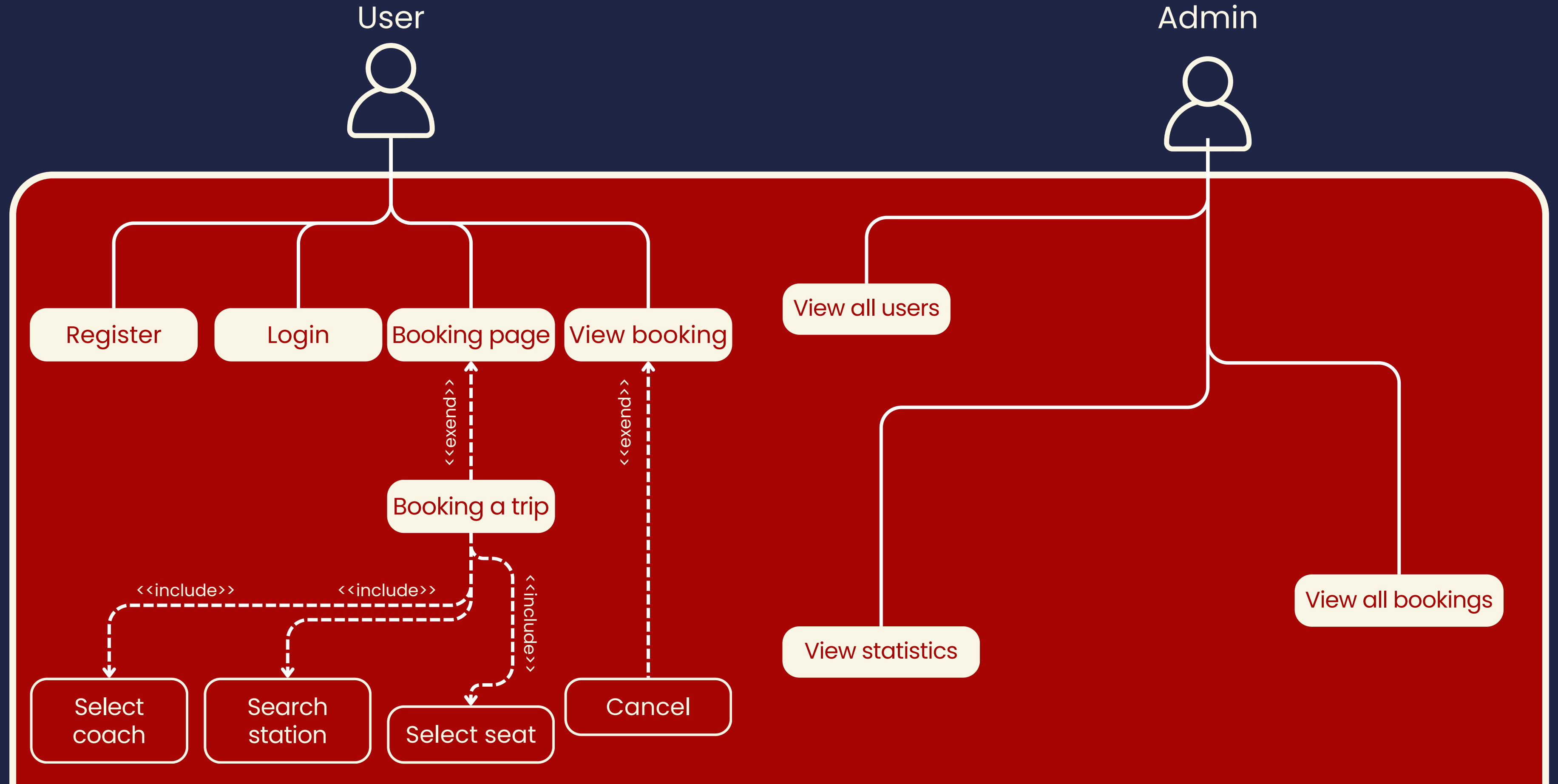
Server



Database



Actors & Main Use Cases



Account Management (Register & Login)

Create Your Account

Username

Choose a username (3-15 charac

Full Name

Enter your full name

Email

Enter your email

Phone Number

Enter your phone number (10-15

Password

Create a password (min. 6 chara

Confirm Password

Re-enter your password

Register

Already have an account? Login here

Register page

- Register with client-side validation.
- Passwords are validated for minimum length and confirmed before submission.
- Login receives backend response and stores current user in localStorage for session state.



**Successful registration redirects to login;
successful login redirects to homepage.**

Login to Your Account

Username

Enter your username

Password

Enter your password

Login

Don't have an account? Register here

Login page

Train and date selection

Available Trains

2 train(s) found from Hanoi to Ho Chi Minh City on 2025-12-24

Train: Reunification Express SE1

SE1

Hanoi 06:00:00



Ho Chi Minh City 10:00:00

Book Now

Train: Reunification Express SE3

SE3

Hanoi 08:00:00



Ho Chi Minh City 12:00:00

Book Now

Select Coach & Seat



Booking page overview

On booking page, selecting a train loads available coaches for the chosen date.



Coach

Choosing a coach triggers loading of available seats; booked seats are disabled/greyed out.



UI

UI shows coach name, available seats count and seat identifiers; user picks one or more seats.



Frontend & Backend

Frontend calls `/coaches/{trainId}` and `/seats/{coachId}` endpoints (or simulated endpoints)

Passenger Details

Number of
Passengers

1

Coach

Coach 3 - 64 seats available



Passenger 1

Full Name

Enter full name

Seat Number

Seat 3A



Payment Method



Credit/Debit Card

Visa, Mastercard, JCB



MoMo

E-Wallet Payment



Bank Transfer

Direct transfer

Card Number

1234 5678 9012 3456

Expiry Date

MM/YY

CVV

123



Payment choice

User fills passenger details for each passenger, selects payment method (simulated Card/MoMo/Bank), and reviews fare breakdown



Booking Reference

Clicking "Confirm & Pay" performs client-side validation then shows a success modal with booking reference (e.g., BKxxxxx).



Booking Summary

Booking summary includes fare, fees, coach and seat numbers and booking reference for records.



Receipt & Email Integration

Backend booking endpoint stores booking record; receipt and email integration planned.

Complete Booking

Manage Bookings



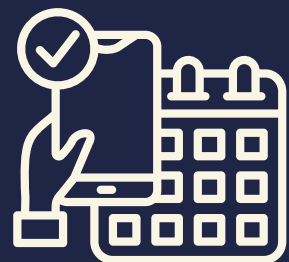
“My Bookings” lists all bookings for the logged-in user with booking reference, train info, coach/seat and status.



Options include viewing details and cancelling bookings if the system allows.



Bookings are stored in database entities and accessible via REST endpoints.



My Bookings

Booking ID: BK784BE505

CONFIRMED

Train Name

Train Fast

Train

TRAIN_01 (TRAIN_01)

Travel Date

23 Dec 2025

From

Ha Noi

To

Ho Chi Minh

Number of Tickets

1

Total Amount

1.550.000 đ

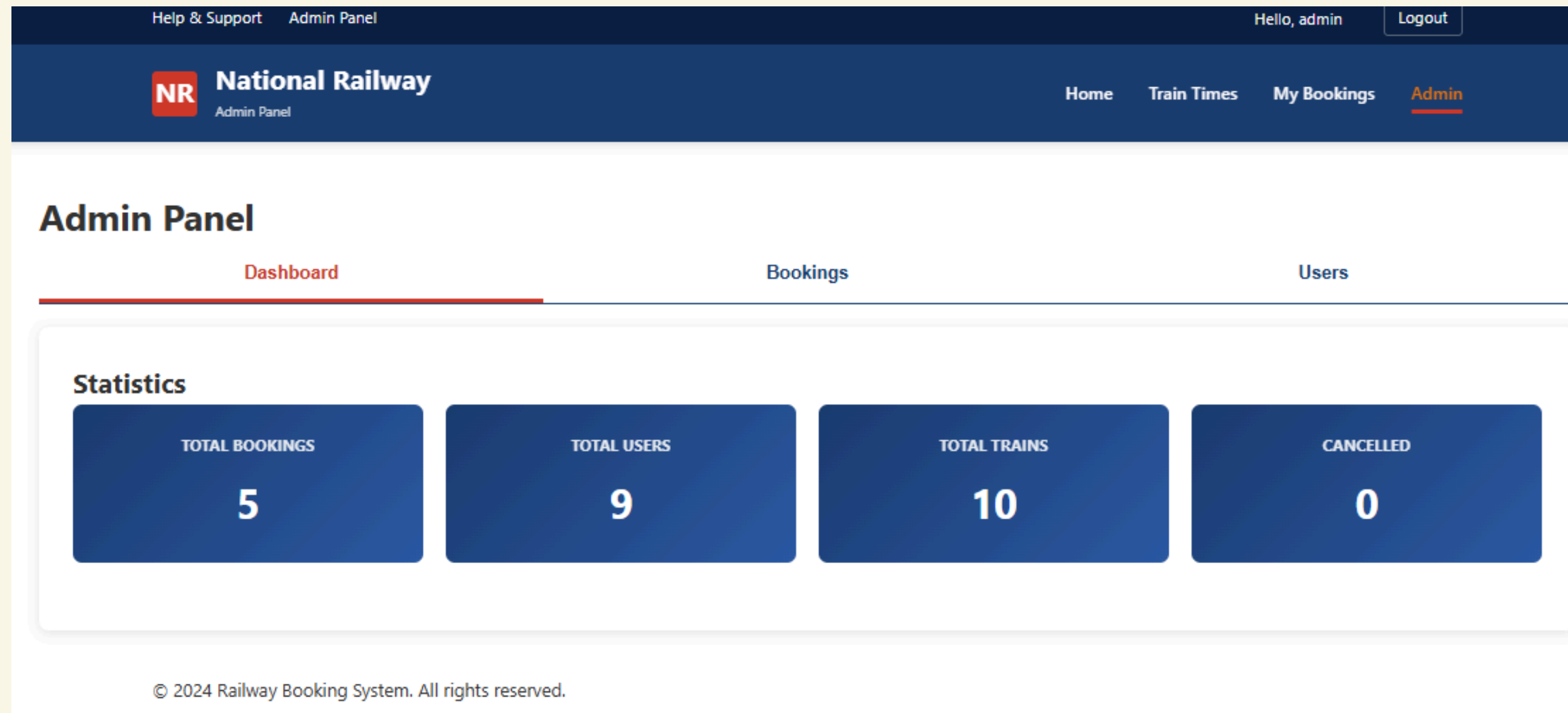
Booking Date

21 Dec 2025

View Details

Cancel Booking

Admin Dashboard



Features: CRUD for Stations, Trains and Schedules; manage coach and seat inventories; user management.



Admin pages included tables with search, pagination.



Security: admin endpoints will require elevated privileges and authentication.

**THANKS FOR
LISTENING &
TIME FOR DEMO**

Q&A